Supplementary Material for "Viability-Preserving Rollout Approximate Dynamic Programming for Required-Target Orbital Inspection"

S1. Required-Target Profile Inventory

Table S1 records the frozen synthetic target identities used in the first required-target campaign. Three equal-width slabs span the longest axis of the transformed mesh. Within each slab, selection begins with the sample nearest the slab centroid and then applies normalized farthest-point sampling. These synthetic profiles distribute prescribed samples across the observable station geometry.

S2. Depth Diagnostic and Timing Protocol

The six-depth diagnostic followed the fixed-library confirmation, for which depth three was already fixed. It used the same 12 validation scenarios with the nine-target requirement, graph, action budget, implementation, and zero-gain connecting actions unchanged. The nine scenarios with required visibility were completed at every depth; the other three lacked an available view of at least one required target. Mean graph cost in Table S2 therefore uses the nine commonly completed scenarios, whereas timing and safety-screen counts include all 12 cases.

Each scenario–depth pair was measured once in ascending-depth order using Python 3.14 on an Apple M4 host running macOS. Complete-route timing includes preliminary base-policy construction but excludes graph loading and serialization. Scheduling and thermal effects were uncontrolled. Safety-screen counts exclude the separate preliminary base-policy call. These timings characterize the computation–cost trade-off of the recorded implementation under these execution conditions.

Depth three reduced mean graph cost by 11.19% relative to depth two. Depth four reduced it by a further 2.74%, at 4.63 times the median planning time and 4.74 times the mean safety-screen count. Depths five and six produced no further cost or completion improvement. These results support depth three as a practical computation–cost compromise; this setting was retained for the subsequent all-solvable benchmark.

S3. Historical Required-Target Campaign

Table S3 preserves the first campaign that motivated a completion-capable comparator. Each profile used different scenario seeds, so cross-profile comparisons are descriptive. The historical local search was initialized with the greedy route; a failed greedy construction therefore left it without a complete incumbent. This differs from the main manuscript's fresh comparator, which initializes local search from a completed one-step approximate dynamic programming route.

The campaign retained all generated cases without feasibility rejection. One shifted case in the six-target profile preserved an available view for every required target but lacked a finite depth-three completion certificate; thus target-wise visibility alone did not establish route feasibility.

S4. Exact Reduced-Graph Bellman Result

For fixed finite nodes, deterministic visibility masks and audited edge costs, a required set $\mathcal{K}$, the no-revisit rule, and horizon H, exhaustive evaluation of the Bellman recursion in the main manuscript returns a minimum-cost task-completing shield-feasible sequence when one exists. It returns $+\infty$ when no such sequence exists in the sampled graph.

To prove the statement, use induction on the remaining budget h. At $h = 0$, the recursion returns zero exactly for a completed state and $+\infty$ otherwise. Assume it is correct for all states with at most $h-1$ remaining actions. Every feasible sequence from a state with budget h begins with one shield-admissible action and a continuation with budget $h - 1$. By the induction hypothesis, the recursion assigns that continuation its minimum cost or $+\infty$ if none exists. Minimizing over every admissible first action therefore returns the minimum total cost, or $+\infty$ when every continuation is infeasible. This establishes exact optimality over the specified finite candidate graph.

S5. ROS 2 Execution Provenance

The selected Robot Operating System 2 (ROS 2) execution in the main manuscript followed an earlier nine-view run that achieved 77.22% weighted background coverage. Development then introduced a 95% coverage goal while retaining the same nine required target IDs and 41-sample weighting. Candidate views 5 and 12 entered the caution zone and were shifted by 8 m and 6 m, respectively. Camera aims were adjusted, and the affected transfers and visibility masks were recomputed before the

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TABLE S1
Frozen synthetic required-target profiles. IDs omit the common mesh prefix; the nine-target profile is primary

Profile	Items	Slab counts	Required sample IDs
6-target	6	2/2/2	00055, 00052, 00007, 00001, 00033, 00015
9-target	9	3/3/3	00055, 00052, 00059, 00007, 00001, 00079, 00033, 00015, 00019
12-target	12	4/4/4	00055, 00052, 00059, 00064, 00007, 00001, 00079, 00045, 00033, 00015, 00019, 00024

TABLE S2
Post-confirmation depth diagnostic on the 12 validation scenarios. Ratios use depth three; negative ΔJ denotes lower mean graph cost

Depth	Complete	Mean J	ΔJ (%)	Median time (s)	Time ($\times$)	Mean screens	Screens ($\times$)
1	9/12	135.714	+14.51	0.0272	0.03	7,132	0.02
2	9/12	133.450	+12.60	0.2207	0.22	65,928	0.20
3	**9/12**	**118.520**	**+0.00**	**1.0094**	**1.00**	**322,349**	**1.00**
4	9/12	115.272	−2.74	4.6760	4.63	1,527,408	4.74
5	9/12	115.272	−2.74	17.6601	17.50	5,757,081	17.86
6	9/12	115.272	−2.74	53.0133	52.52	17,454,611	54.15

TABLE S3
Historical completion counts from the first required-target campaign

Profile	Split	Depth-3 ADP	Greedy-seeded local
6-target	Test	40/50	0/50
9-target	Test	37/50	0/50
12-target	Test	38/50	0/50
6-target	Shifted	10/30	0/30
9-target	Shifted	10/30	0/30
12-target	Shifted	10/30	1/30

reported execution. Each 90 s transfer was followed by 60 s of settling.

The run used Ubuntu 24.04.4, ROS 2 Jazzy, Gazebo Harmonic, RViz2, and Python 3.12.3. Its logs contain 36,172 safe-control messages, including 1,847 safety-filter interventions, and 36,181 trajectory samples. A $247\,525$-triangle full-mesh audit was applied to the swept trajectory. The maximum publisher gap was 0.249 s, compared with the nominal 0.075 s diagnostic interval. Recorded camera frames matched their observation-event receipts within 0.074 s. Both recorded bags contain identical verification events, and Gazebo poses confirm visual alignment with the ROS state.

The Gazebo visualization uses an RGB perspective camera with a 70° horizontal field of view (FOV), 960×686 pixels, and a 15-Hz update rate. Near and far clipping distances are 0.1 and 120 m. The image aspect ratio implies an approximately 53.2° vertical FOV, whereas target-credit checks use the planner's 50° vertical FOV. The camera is offset 0.45 m along the chaser body forward axis. Gazebo images are bridged to ROS 2 as `sensor_msgs/msg/Image` on `/chaser/camera/image` for display and recording. Target credit is computed from geometric visibility and observation-acceptance gates; rendered images support visualization and recording.

These records document the viewpoint refinements, safety-filter interventions, and synchronization of the reported development execution.